

ECE3411

Lecture 13

Operational Amplifier (Op-Amp)

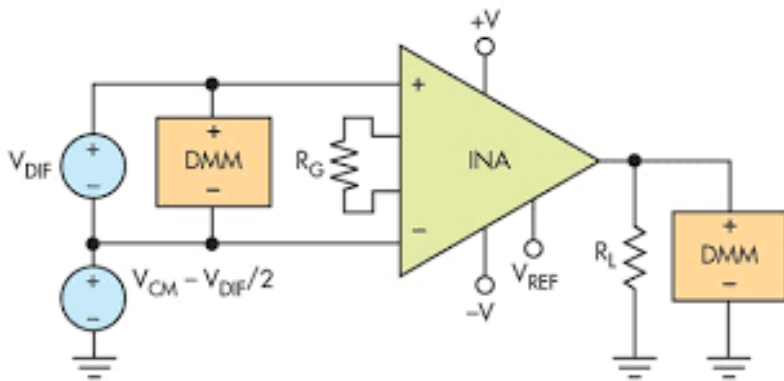
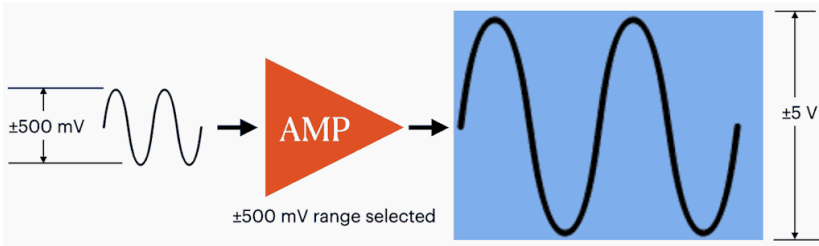
Sung Yeul Park

Department of Electrical & Computer Engineering

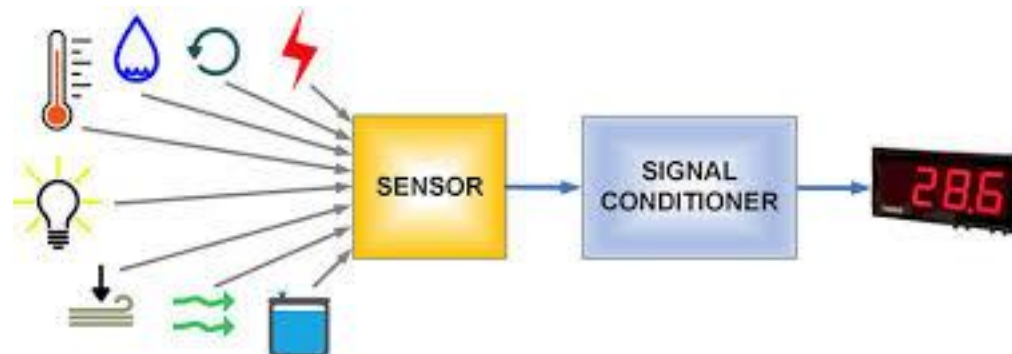
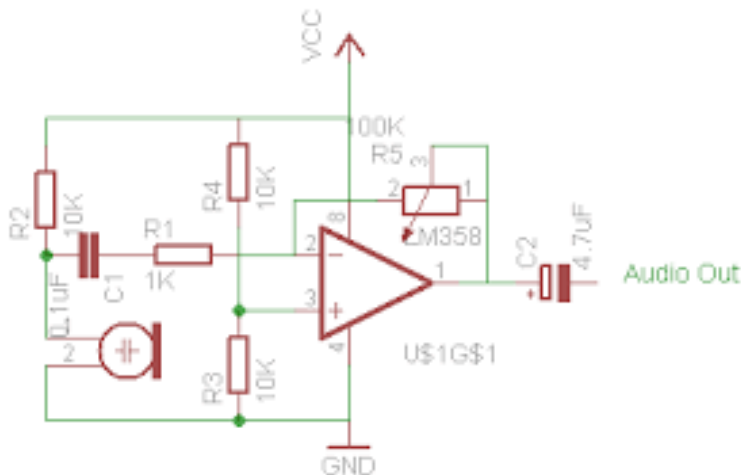
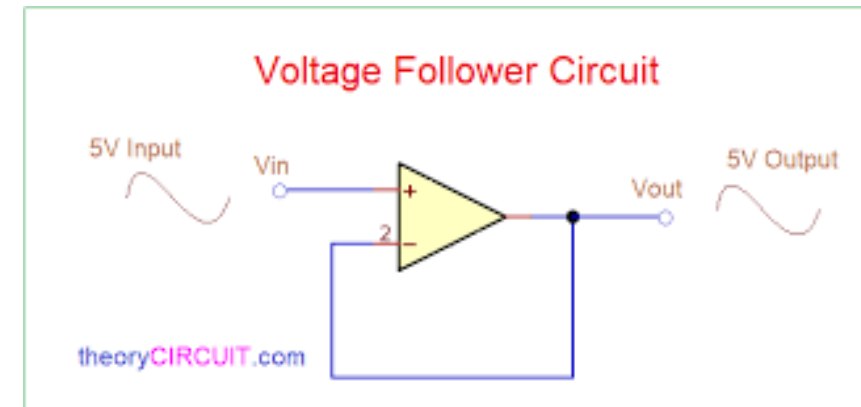
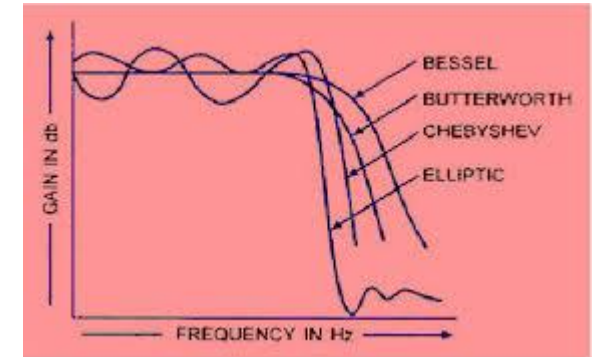
University of Connecticut

Email: sung_yeul.park@uconn.edu

Op-Amp Applications



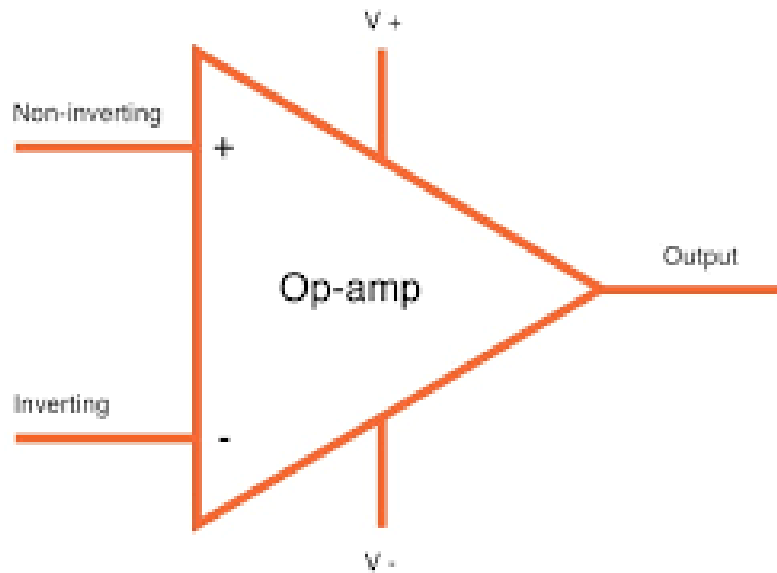
- Signal amplification
- Active filters
- Instrumentation amplifiers
- Voltage followers/buffers
- Comparator circuits
- Audio pre-amplifiers
- Sensor signal conditioning
- Current sensing circuits



Introduction

■ Overview

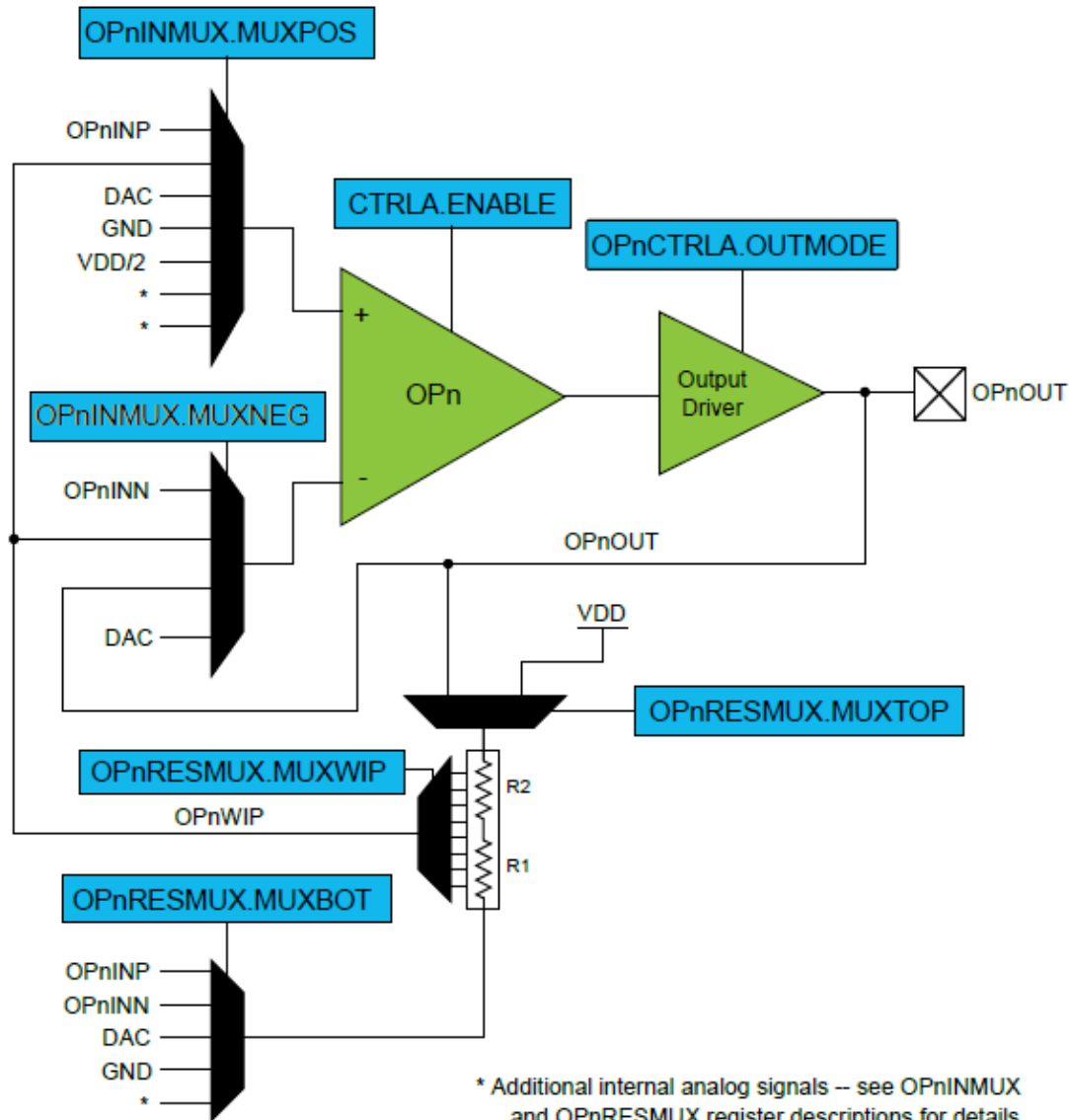
The AVR128DB48 features three built-in operational amplifiers (OPAMP0, OPAMP1, OPAMP2) that provide analog signal conditioning and amplification capabilities. These op-amps are integrated into the microcontroller, eliminating the need for external analog components in many applications.



Key Features

- Three independent op-amps (OPAMP0, OPAMP1, OPAMP2)
- Rail-to-rail input and output operation
- Programmable gain amplifier (PGA) configurations
- Internal feedback networks for common gain settings
- Low offset voltage and low noise
- Wide bandwidth suitable for audio and control applications
- Low power consumption with shutdown capability
- Multiple input/output routing options

Diagram of Op-Amp



* Additional internal analog signals -- see OPnINMUX and OPnRESMUX register descriptions for details

Pin Configuration

Op-Amp	Positive Input	Negative Input	Output
Op-Amp 0	PD1 (AIN1)	PD3 (AIN3)	PD2 (AIN2)
Op-Amp 1	PD4 (AIN4)	PD7 (AIN5)	PD5 (AIN6)
Op-Amp 2	PE1 (AIN7)	PE3 (AIN8)	PE2 (AIN9)

Initialization of Op-Amp

1. Configure the timebase of the OPAMP peripheral in OPAMP.TIMEBASE
2. For each op amp that will be used:
 - a. Configure the Op Amp n Input Multiplexer (OPAMP.OPnINMUX):
 - Select the non-inverting (+) input of the op amp and the inverting (-) input of the op amp
 - b. Configure the Op Amp n Resistor Ladder Multiplexer (OPAMP.OPnRESMUX):
 - Select the connection to the top resistor, the bottom resistor, and the wiper position in the resistor ladder
 - c. Configure the Op Amp n Control A (OPAMP.OPnCTRLA) register:
 - Configure the op amp to be always on, • Disable events for the op amp, and • Select the normal output mode
 - d. Configure the allowed settling time
3. Enable the OPAMP peripheral
4. For each op amp whose settling time was configured in step 2.4 above, wait for the SETTLED bit in the Op Amp n Status (OPAMP.OPnSTATUS) register to become '1'. This indicates that the op amp start-up and settling have completed, and the OPAMP peripheral is ready for use.

MUXPOS/MUXNEG – Non-Inverting (+)/Inverting (-) Input

OPn has the following additional inputs available:

- INP (OPnINP) - Positive input pin for OPn
- WIP (OPnWIP) - Wiper from OPn's resistor ladder
- DAC - DAC output

(DAC and DAC output buffer must be enabled)

- GND - Ground
- VDDDIV2 - VDD/2

OP1 has the following additional input available:

- LINKOUT (OP0OUT) - OP0 output

OP2 has the following additional inputs available:

- LINKOUT (OP1OUT) - OP1 output
- LINKWIP (OP0WIP) - Wiper from OP0's resistor ladder

=====

OPn:

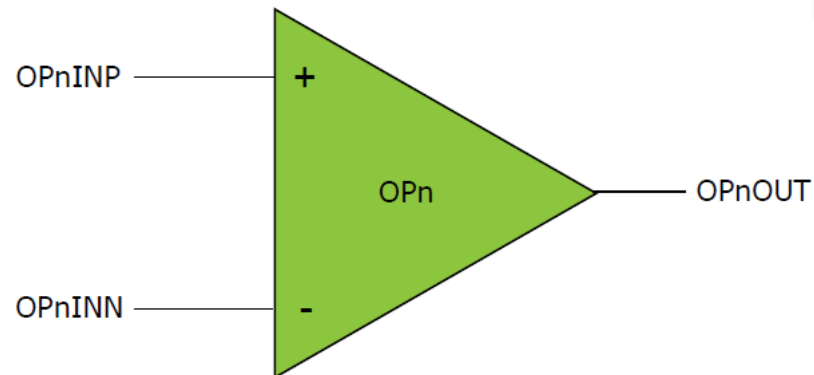
- INN (OPnINN) - Negative input pin for OPn
- WIP (OPnWIP) - Wiper from OPn's resistor ladder
- OUT (OPnOUT) - OPn output
- DAC - DAC output (DAC and DAC output buffer must be enabled)

TOP, BOT, WIP Connection

TOP	BOT	WIP
• OFF - Multiplexer off, no signal connected • OUT (OPnOUT) - OPn output • VDD - VDD	• OFF - Multiplexer off, no signal connected • INP (OPnINP) - Positive input pin for OPn • INN (OPnINN) - Negative input pin for OPn • DAC - DAC output (DAC and DAC output buffer must be enabled) • GND - Ground	• WIP0 - $R2/R1 = 1/15$ • WIP1 - $R2/R1 = 1/7$ • WIP2 - $R2/R1 = 1/3$ • WIP3 - $R2/R1 = 1$ • WIP4 - $R2/R1 = 5/3$ • WIP5 - $R2/R1 = 3$ • WIP6 - $R2/R1 = 7$ • WIP7 - $R2/R1 = 15$

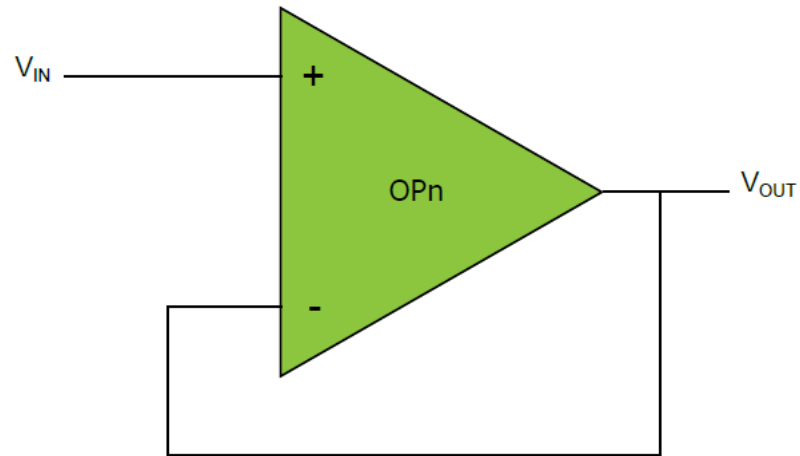
Op Amp Connected Directly to Pins

	OPnINMUX		OPnRESMUX		
	MUXPOS	MUXNEG	MUXBOT	MUXWIP	MUXTOP
OPn	INP	INN	OFF	WIPO	OFF



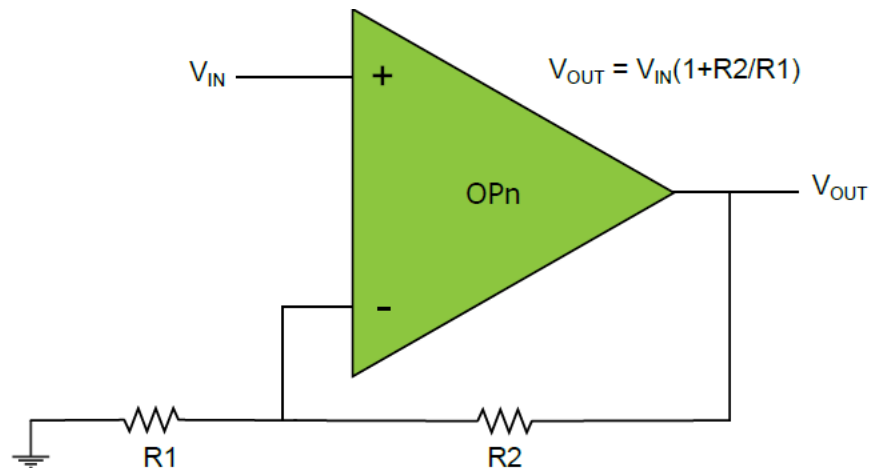
Op Amp Connected Voltage Follower

	OPnINMUX		OPnRESMUX		
	MUXPOS	MUXNEG	MUXBOT	MUXWIP	MUXTOP
OPn	INP	OUT	OFF	WIPO	OFF



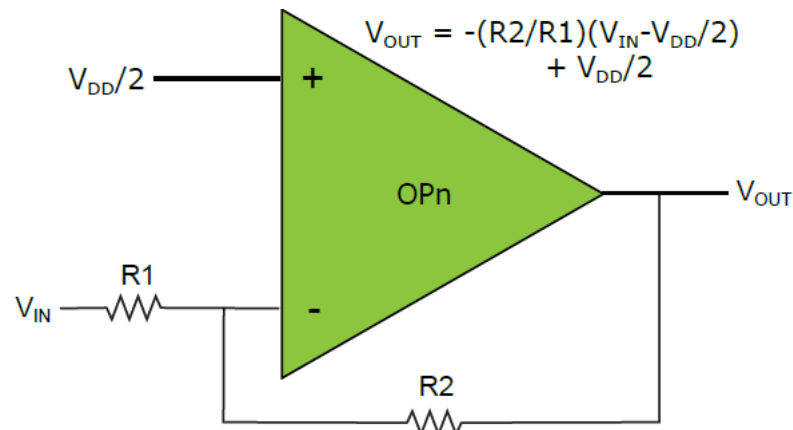
Op Amp Connected Non-Inverting Programmable Gain Amp

	OPnINMUX		OPnRESMUX		
	MUXPOS	MUXNEG	MUXBOT	MUXWIP	MUXTOP
OPn	INP	WIP	GND	Setting determines Gain	OUT



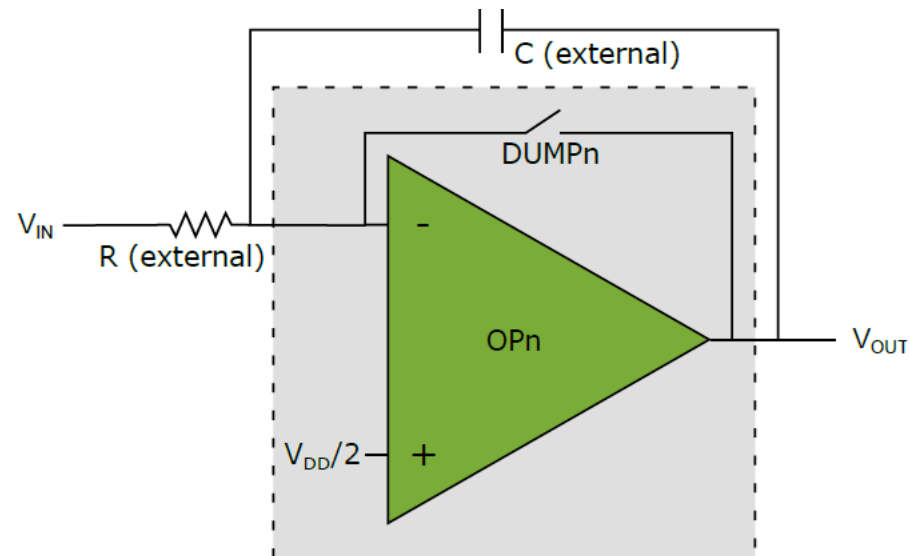
Op Amp Connected Inverting Programmable Gain Amp

	OPnINMUX		OPnRESMUX		
	MUXPOS	MUXNEG	MUXBOT	MUXWIP	MUXTOP
OPn	VDDDIV2	WIP	INN	Setting determines Gain	OUT



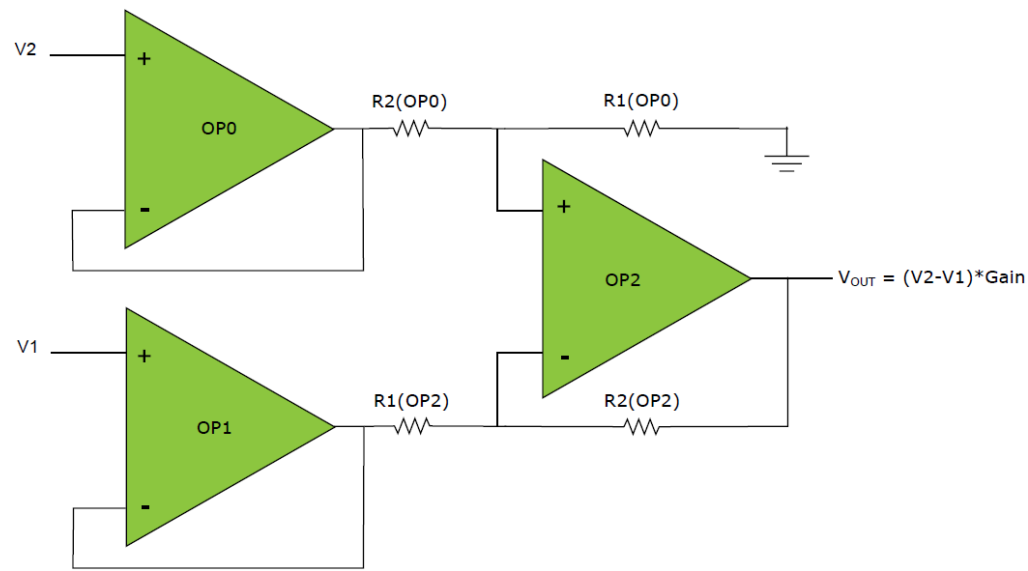
Op Amp Connected Integrator

	OPnINMUX		OPnRESMUX		
	MUXPOS	MUXNEG	MUXBOT	MUXWIP	MUXTOP
OPn	VDDDIV2	INN	OFF	WIPO	OFF



Op Amp Connected Instrumentation Amplifier

	OPnINMUX		OPnRESMUX		
	MUXPOS	MUXNEG	MUXBOT	MUXWIP	MUXTOP
OP0	INP	OUT	GND	See the table	OUT
OP1	INP	OUT	OFF	WIP0	OFF
OP2	LINKWIP (OP0WIP)	WIP	LINKOUT (OP1OUT)	See the table	OUT



Gain Selection for Instrumentation Amplifier

Gain	OP0RESMUX.MUXWIP	OP2RESMUX.MUXWIP
1/15	0x7	0x0
1/7	0x6	0x1
1/3	0x5	0x2
1	0x3	0x3
3	0x2	0x5
7	0x1	0x6
15	0x0	0x7

Key Registers

1. OPAMPn.CTRLA - Control Register A (n = 0, 1, 2)
2. OPAMPn.OP - Op-Amp Control Register
3. OPAMPn.MUXPOS - Positive Input Mux
4. OPAMPn.MUXNEG - Negative Input Mux
5. OPAMPn.MUXOUT - Output Mux
6. OPAMPn.RES - Resistor Ladder Control

OPAMPn.CTRLA - Control Register A

Bit	7	6	5	4	3	2	1	0
								ENABLE
Access								R/W
Reset								0

Bit 0 – ENABLE Enable OPAMP Peripheral

This bit controls whether the OPAMP peripheral is enabled or not.

Value	Description
0	The OPAMP peripheral is disabled
1	The OPAMP peripheral is enabled

Timebase

Bit	7	6	5	4	3	2	1	0
	TIMEBASE[6:0]							
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	1

Bits 6:0 – TIMEBASE[6:0] Timebase

This bit field controls the maximum value of a counter that counts CLK_PER cycles to achieve a time interval equal to or larger than 1 μ s. It should be written with one less than the number of CLK_PER cycles that are equal to or larger than 1 μ s. This is used for the internal timing of the warmup and settling times.

Power Control

Bit	7	6	5	4	3	2	1	0
								IRSEL
Access								R/W
Reset								0

Bit 0 – IRSEL Input Range Select

This bit selects the op amp input voltage range.

Value	Description
0	The op amp input voltage range is rail-to-rail
1	The op amp input voltage range and power consumption are reduced. See the <i>Electrical Characteristics</i> section for more information.

Op Amp n CTRLA: OPnCTRLA

Bit	7	6	5	4	3	2	1	0
	RUNSTDBY				OUTMODE[1:0]		EVENTEN	ALWAYSON
Access	R/W				R/W		R/W	R/W
Reset	0				0		0	0

Bit 7 – RUNSTDBY Run in Standby Mode

This bit controls whether or not the op amp functions in Standby sleep mode.

Value	Description
0	OPn is disabled when in Standby sleep mode, and its output driver is disabled
1	OPn will continue operating as configured in Standby sleep mode

Bits 3:2 – OUTMODE[1:0] Output Mode

This bit field selects the output mode of the output driver.

Value	Name	Description
0x0	OFF	The output driver for OPn is disabled, but this can be overridden by the DRIVEN event
0x1	NORMAL	The output driver for OPn is enabled in Normal mode
0x2 – 0x3	-	Reserved

Bit 1 – EVENTEN Event Enable

This bit enables event reception and generation.

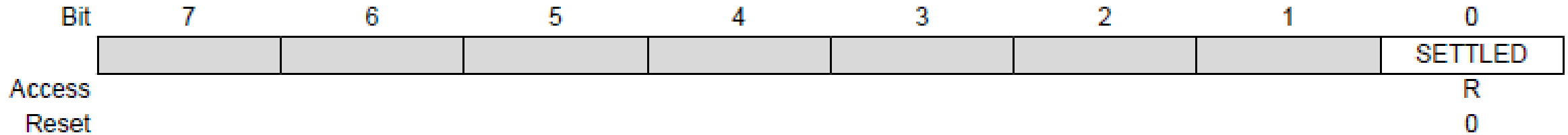
Value	Description
0	No events are enabled for OPn
1	All events are enabled for OPn

Bit 0 – ALWAYSON Always On

This bit controls whether the op amp is always on or not.

Value	Description
0	OPn is not always on but can be enabled by the ENABLEn event and disabled by the DISABLEn event
1	OPn is always on

Op Amp n Status: OPnSTATUS



Bit 0 – SETTLED Op Amp has Settled

This bit is cleared when the op amp is waiting for settling related to enabling or configuration changes.

This bit is set when the allowed settling time is finished.

Op Amp n Resistor Ladder Multiplexer: OPnRESMUX

Bit	7	6	5	4	3	2	1	0
	MUXWIP[2:0]			MUXBOT[2:0]			MUXTOP[1:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 7:5 – MUXWIP[2:0] Multiplexer for Wiper

This bit field selects the resistor ladder wiper (potentiometer) position.

Value	Name	Description
0x0	WIP0	R1 = 15R, R2 = 1R
0x1	WIP1	R1 = 14R, R2 = 2R
0x2	WIP2	R1 = 12R, R2 = 4R
0x3	WIP3	R1 = 8R, R2 = 8R
0x4	WIP4	R1 = 6R, R2 = 10R
0x5	WIP5	R1 = 4R, R2 = 12R
0x6	WIP6	R1 = 2R, R2 = 14R
0x7	WIP7	R1 = 1R, R2 = 15R

Op Amp n Resistor Ladder Multiplexer: OPnRESMUX

Bit	7	6	5	4	3	2	1	0
	MUXWIP[2:0]			MUXBOT[2:0]			MUXTOP[1:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 4:2 – MUXBOT[2:0] Multiplexer for Bottom

This bit field selects the analog signal connected to the bottom resistor in the resistor ladder.

Value	Name	Description
0x0	OFF	Multiplexer off
0x1	INP	Positive input pin for OPn
0x2	INN	Negative input pin for OPn
0x3	DAC	DAC output (DAC and DAC output buffer must be enabled)
0x4	LINKOUT	OP[n-1] output ⁽¹⁾
0x5	GND	Ground
Other	-	Reserved

Note:

1. When selecting LINKOUT for OP0, MUXBOT is connected to the output of OP2.

Bits 1:0 – MUXTOP[1:0] Multiplexer for Top

This bit field selects the analog signal connected to the top resistor in the resistor ladder.

Value	Name	Description
0x0	OFF	Multiplexer off
0x1	OUT	OPn output
0x2	VDD	V _{DD}
Other	-	Reserved

Op Amp n Input Multiplexer: OPnINMUX

Bit	7	6	5	4	3	2	1	0
		MUXNEG[2:0]				MUXPOS[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		0	0	0		0	0	0

Bits 6:4 – MUXNEG[2:0] Multiplexer for Negative Input

This bit field selects which analog signal is connected to the inverting (-) input of OPn.

Value	Name	Description
0x0	INN	Negative input pin for OPn
0x1	WIP	Wiper from OPn's resistor ladder
0x2	OUT	OPn output (unity gain)
0x3	DAC	DAC output (DAC and DAC output buffer must be enabled)
Other	-	Reserved

Bits 2:0 – MUXPOS[2:0] Multiplexer for Positive Input

This bit field selects which analog signal is connected to the non-inverting (+) input of OPn.

Value	Name	Description
0x0	INP	Positive input pin for OPn
0x1	WIP	Wiper from OPn's resistor ladder
0x2	DAC	DAC output (DAC and DAC output buffer must be enabled)
0x3	GND	Ground
0x4	VDDDIV2	$V_{DD}/2$
0x5	LINKOUT	OP[n-1] output (Setting only available for OP1 and OP2)
0x6	LINKWIP	Wiper from OP0's resistor ladder (Setting only available for OP2)
Other	-	Reserved

Op Amp n Settle Timer

Bit	7	6	5	4	3	2	1	0
	SETTLE[6:0]							
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bits 6:0 – SETTLE[6:0] Settle Timer

This bit field specifies the number of microseconds allowed for the op amp output to settle. This value, together with the value in the TIMEBASE register, is used by an internal timer to determine when to generate the READYn event and set the SETTLED flag in the OPnSTATUS register.

Op Amp n Calibration

Bit	7	6	5	4	3	2	1	0
	CAL[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	x	x	x	x	x	x	x	x

Bits 7:0 – CAL[7:0] Calibration of Input Offset Voltage

This bit field is a calibration value that adjusts the input offset voltage of the op amp. '0x00' provides the most negative value of offset adjustment, '0x80' provides no offset adjustment, and '0xFF' provides the most positive value of offset adjustment.

Initialization of Op-Amp

1. Configure the timebase of the OPAMP peripheral in OPAMP.TIMEBASE
2. For each op amp that will be used:
 - a. Configure the Op Amp n Input Multiplexer (OPAMP.OPnINMUX):
 - Select the non-inverting (+) input of the op amp and the inverting (-) input of the op amp
 - b. Configure the Op Amp n Resistor Ladder Multiplexer (OPAMP.OPnRESMUX):
 - Select the connection to the top resistor, the bottom resistor, and the wiper position in the resistor ladder
 - c. Configure the Op Amp n Control A (OPAMP.OPnCTRLA) register:
 - Configure the op amp to be always on, • Disable events for the op amp, and • Select the normal output mode
 - d. Configure the allowed settling time
3. Enable the OPAMP peripheral
4. For each op amp whose settling time was configured in step 2.4 above, wait for the SETTLED bit in the Op Amp n Status (OPAMP.OPnSTATUS) register to become '1'. This indicates that the op amp start-up and settling have completed, and the OPAMP peripheral is ready for use.

Microchip Op-amp Training

<https://github.com/microchip-pic-avr-examples/avr128db48-getting-started-with-opamp-studio>

- **Hardware Prerequisites**

- The Analog Signal Conditioning (OPAMP) peripheral features up to three internal operational amplifiers (op amps). The op amps can be configured to a multitude of different operations using internal multiplexers and resistor ladders
- The main purpose of op amps is to condition the analog signals before acquisition in a microcontroller or to provide the necessary output drive in control applications
- These examples show the following configurations of the OPAMP peripheral on the AVR DB family of microcontrollers:
 - **Simple Op Amp:** OP0 is connected directly to the pins
 - **Voltage Follower:** OP0 is configured as a voltage follower
 - **Non-Inverting PGA:** OP0 is configured as a non-inverting PGA
 - **Differential Amplifier:** OP0 and OP1 are combined to create a differential amplifier
 - **Instrumentation Amplifier:** OP0, OP1 and OP2 are combined to create an instrumentation amplifier

Voltage follower(1)

- In this assignment, the AVR128DB48 Curiosity Nano will be configured to use the op amp OP0 in the Analog Signal Conditioning (OPAMP) peripheral as a voltage follower.
- The Digital-to-Analog Converter (DAC) is configured to create a sinusoidal signal, which will be used as an input to the op amp.
- The Analog-to-Digital Converter (ADC) is configured to sample the output of the op amp.
- Both the input and output of the op amp are streamed over to the MPLAB® Data Visualizer using the data streamer protocol.
- The starting point for this assignment is the MPLAB® X project *Assignment3.X*.
- **Objectives**
 - Understand the necessary steps to configure an op amp instance in the OPAMP peripheral
 - Configure OP0 as a voltage follower
 - Plot the input and output from OP0 using the MPLAB® Data Visualizer

Voltage follower(2)

- Configure OP0 as a Voltage Follower
- Edit the OPAMP0_init() so it configures OP0 to be a voltage follower
- Plot the input and output of OP0 in the MPLAB® Data Visualizer

- 1. Edit OPAMP0_init() to set the time base for the OPAMP peripheral by writing:

```
OPAMP.TIMEBASE = OPAMP_TIMEBASE_US;
```

```
#define OPAMP_TIMEBASE_US (ceil(F_CPU / 1e6) - 1)
```

- 2. Set the output mode to normal and the Always On (ALWAYSON) bit for OP0 by writing:

```
OPAMP.OP0CTRLA = OPAMP_OP0CTRLA_OUTMODE_NORMAL_gc | OPAMP_ALWAYSON_bm;
```

- 3. Configure OP0 to be a voltage follower with the DAC as an input by writing:

```
OPAMP.OP0INMUX = OPAMP_OP0INMUX_MUXNEG_OUT_gc | OPAMP_OP0INMUX_MUXPOS_DAC_gc;
```

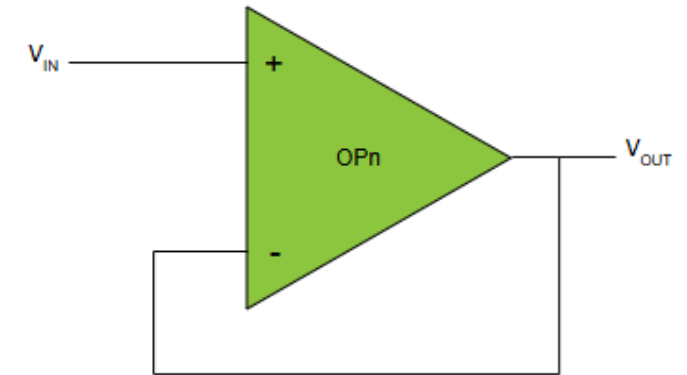
```
/* Configure the Op Amp n Resistor Wiper Multiplexer */
```

```
OPAMP.OP0RESMUX = OPAMP_OP0RESMUX_MUXBOT_OFF_gc |
```

```
OPAMP_OP0RESMUX_MUXWIP_WIPO_gc | OPAMP_OP0RESMUX_MUXTOP_OFF_gc;
```

Voltage follower(3)

Figure 5-1. Op Amp in Voltage Follower Configuration



- 4. Set the settling time by writing:

```
OPAMP.OPOSETTL = OPAMP_MAX_SETTLE;
```

- 5. Enable the OPAMP peripheral by writing:

```
OPAMP.OPOCTRLA = OPAMP_OPOCTRLA_OUTMODE_NORMAL_gc | OPAMP_ALWAYS_ON_bm;
```

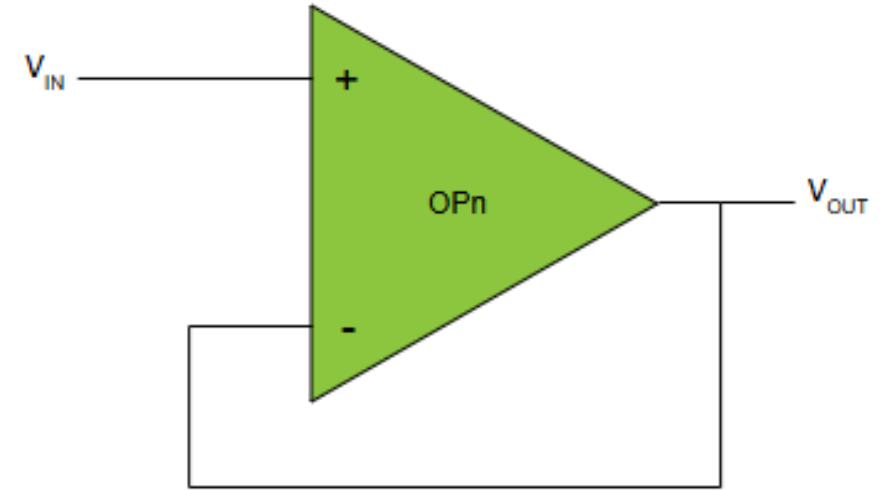
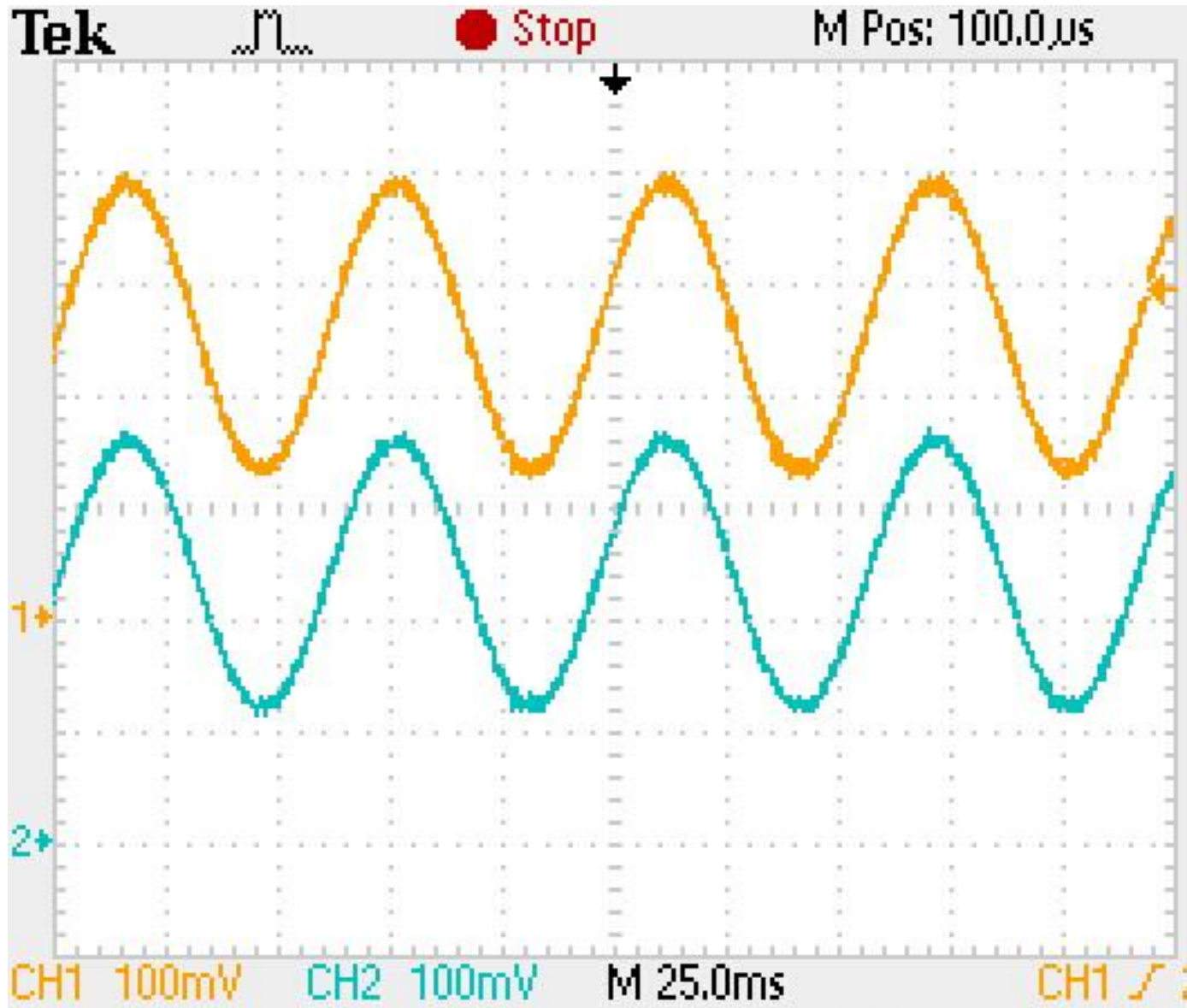
- 6. Wait for the op amp to settle before leaving the initialization by checking the Op Amp has Settled (SETTLED) bit field in the Op Amp n Status (OPnSTATUS) register:

```
while (!(OPAMP.OPnSTATUS & OPAMP_SETTLED_bm))
```

```
{  
    ;  
};
```

- 7. Verify that the solution/project builds by selecting the *Build Main Project* from the top menu bar in MPLAB® X or by pressing the *F11* key.
- 8. Flash the device by selecting the *Make and Program Device Main Project* from the top menu bar in MPLAB® X.

Voltage follower(4)



Channel 1: OP0 Output (PD2)
Channel 2: DAC0 Output (PD6)

Ch1 and Ch2 are in phase and
the same magnitude

Sample Code(1)

```
#include <avr/io.h>
#include <avr/cpufunc.h>
#include "src/rtc.h"
#include "src/adc.h"
#include "src/dac.h"
#include "src/uart.h"

/*Configure system fuses ensuring known start
condition for MCU*/
void GPIO_init(void);
void OPAMP0_init(void);

void system_init(void) {
    CPU_CCP = CCP_IOREG_gc;
    CLKCTRL.XOSCHFCTRLA = CLKCTRL_FRQRANGE_16M_gc |
    CLKCTRL_ENABLE_bm;
    CPU_CCP = CCP_IOREG_gc;
    CLKCTRL.MCLKCTRLA = CLKCTRL_CLKSEL_EXTCLK_gc;
    // Wait for clock to stabilize
    _delay_ms(100);
}

int main(void)
{
    /* Set Main Clock to 24 MHz */
    system_init();

    GPIO_init();
```

```
    LED0_init();

    RTC_init();    /*1s interrupt timer*/

    DAC0_init();/*Creates and outputs the sine waveform*/

    USART_init();    /*Streams data into Data Visualizer*/

    ADC0_init();    /*Acquire Signal from OPAMP0 output*/

    /*Start the timer controlling the sine waveform
    and the timer controlling the acquisition*/
    DAC0_SineWaveTimer_enable();
    ADC0_SampleTimer_enable();

    OPAMP0_init();

    RTC_enable();

    sei();    /*Enable global interrupts*/

    while (1)
    {
        ;
    }
}
```


Sample Code(2)

```
#define OPAMP_TIMEBASE_US      (ceil(F_CPU /1e6)-1)
#define OPAMP_MAX_SETTLE      (0x7F)

void OPAMP0_init (void)
{
    /* Configure the Timebase */
    OPAMP.TIMEBASE = OPAMP_TIMEBASE_US;

    /* Configure the Op Amp n Control A */
    OPAMP.OP0CTRLA = OPAMP_OP0CTRLA_OUTMODE_NORMAL_gc
| OPAMP_ALWAYS_ON_bm;

    /* Configure the Op Amp n Input Multiplexer */
    OPAMP.OP0INMUX = OPAMP_OP0INMUX_MUXNEG_OUT_gc |
OPAMP_OP0INMUX_MUXPOS_DAC_gc;

    /* Configure the Op Amp n Resistor Wiper
Multiplexer */
    OPAMP.OP0RESMUX = OPAMP_OP0RESMUX_MUXBOT_OFF_gc |
OPAMP_OP0RESMUX_MUXWIP_WIP0_gc |
OPAMP_OP0RESMUX_MUXTOP_OFF_gc;

    /* Configure the Op Amp n Settle Time*/
    OPAMP.OP0SETTLE = OPAMP_MAX_SETTLE;

    /* Enable OPAMP peripheral */
    OPAMP.CTRLA = OPAMP_ENABLE_bm;

    /* Wait for the operational amplifiers to settle */
    while (!(OPAMP.OP0STATUS & OPAMP_SETTLED_bm))
    {
        ;
    }
}

void GPIO_init(void)
{
    /* Disable digital input buffer*/
    PORTD.PIN1CTRL = PORT_ISC_INPUT_DISABLE_gc;
    PORTD.PIN3CTRL = PORT_ISC_INPUT_DISABLE_gc;
}
```

Lab Practice #12: Shunt Current Sensing

- Students will use the AVR128DB48's internal op-amps to build a 3-op-amp instrumentation amplifier to measure the voltage drop across a shunt resistor, which corresponds to the current flowing through a load.
- Connect OP2 output to ADC0 and display the current value every 2 sec using USART.

Materials Required:

- 0.1-ohm shunt resistor
- Bench Power supply (e.g., 5V regulated)
- Load resistor (e.g., 100Ω)
- Breadboard and jumper wires
- Serial terminal (PuTTY)

